



CVM
UNIVERSITY

Aegis: Charutar Vidya Mandal (Estd.1945)

FACULTY OF ENGINEERING & TECHNOLOGY

Effective from Academic Batch: 2022-23

Programme: Bachelor of Technology (Computer Science and Engineering (IoT))

Semester: VI

Course Code: 202046714

Course Title: IoT Architecture and Protocols

Course Group: Professional Core Course

Course Objectives: The main objective of the course is to make students know the basic concept and architecture of embedded systems and different design protocols used for an embedded system for IoT applications. Students will gain in depth knowledge about the IoT enabled technology.

Teaching & Examination Scheme:

Contact hours per week			Course Credits	Examination Marks (Maximum / Passing)				
Lecture	Tutorial	Practical		Theory		J/V/P*		Total
				Internal	External	Internal	External	
3	0	2	4	50/18	50/17	25/9	25/9	150/53

* J: Jury; V: Viva; P: Practical

Detailed Syllabus:

Sr.	Contents	Hours
1	Overview: IoT-An Architectural Overview- Building an Architecture - Main Design Principles And Needed Capabilities - An Iot Architecture Outline - Standards Considerations - Local And Wide Area Networking - Data Management - Business Processes in IoT - Everything as a Service (XaaS).	08
2	IoT Architecture: Drivers Behind New Network Architectures, oneM2M IoT Architecture, IoT World Forum (IoTWF) Architecture, Simplified IoT Architecture.	06
3	IoT Physical, Data link & Network Layer Protocols: Physical/MAC Layer: IEEE 802.11, IEEE 802.15, 3GPP, Wireless HART, Z-wave, Bluetooth, Zigbee, DASH7. Network Layer: IPv4/IPv6, 6LoWPAN, 6TiSCH, ND, DHCP, ICMP, RPL, CORPL, CARP.	08
4	Transport and Session Layer Protocol: Transport Layer: TCP, MPTCP, UDP, DCCP, SCTP, TLS, DTLS. Session Layer: HTTP, CoAP, XMPP, AMPQ, MQTT.	07
5	Service Layer Protocol and Security: Service Level: ETSIM2M, OMA, BBF. Security in IoT Protocols: MAC 802.15.4, 6LoWAN, RPL, Application Layer.	07
6	Case Study and Real-World Applications: Smart City Application, Asset Management, Industrial Automation and Smart Agriculture, IoT Interrogation with UAV.	04
	Total	40



CVM UNIVERSITY

Aegis: Charutar Vidya Mandal (Estd.1945)

List of Practicals / Tutorials:

1	To connect NodeMCU to WiFi.
2	To implement Soft API on NodeMCU (IOT node).
3	To Create a standalone web server that controls outputs (two LEDs).
4	To build a web server with a slider to control the LED bright-ness.
5	To create an SMS notification system that sends an SMS when sensor readings are above or below a certain threshold.
6	To make HTTP GET requests to get values and URL encoded requests with IOT node.
7	To implement client-server communication between two IOT nodes.
8	To make HTTP POST requests to post JSON data (DHT11) to Thing Speak.
9	To send HTTP GET Request from NodeMCU ESP8266 to Thing Speak Cloud.
10	To implement Mosquitto MQTT Broker. To control LED and Switch using MQTT.
11	Project

Reference Books:

1	Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stephan Avesand, Stamatis Karnuskos, David Boyle, "From Machine to Machine to the Internet of Things : Introduction to a New Age of Intelligence", 1 st Edition, Academic Press, 2014.
2	Simone Cirani, Gianluigi Ferrari, Marco Picone, Luca Veltri, Internet of Things, Architecture Protocols and Standards, Willey Publication, 2019.
3	Raj Kamal, Internet of Things: Architecture and Design Principles, McGraw Hill Education.
4	Bern Scholz-Reiter, Florian Michahelles, "Architecting the Internet of Things".
5	Sudip Mishra et al, Introduction to IoT, Cambridge University Press.
6	Hanes et al, IoT Fundamentals, Cisco Press

Supplementary learning Material:

1	NPTEL Video lectures
2	Research papers from reputed Journals (IEEE, Elsevier, Springer etc.)

Pedagogy:

- Direct classroom teaching
- Audio Visual presentations/demonstrations
- Assignments/Quiz
- Continuous assessment
- Interactive methods
- Seminar/Poster Presentation
- Industrial/ Field visits
- Course Projects

Suggested Specification table with Marks (Theory) (Revised Bloom's Taxonomy):

Distribution of Theory Marks in %						R: Remembering; U: Understanding; A: Applying; N: Analyzing; E: Evaluating; C: Creating
R	U	A	N	E	C	
10%	20%	20%	20%	15%	15%	

Note: This specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary slightly from above table.



CVM
UNIVERSITY

Aegis: Charutar Vidya Mandal (Estd.1945)

Course Outcomes (CO):

Sr.	Course Outcome Statements	%weightage
CO-1	To make the students Acquainted about the IoT Architecture.	30
CO-2	Develop a fundamental understanding of IoT protocols at various level and visualize their applications.	30
CO-3	To understand and develop Protocol security while designing IoT applications	20
CO-4	To study the applications of IoT for various field with their use cases.	20

Curriculum Revision:	
Version:	2.0
Drafted on (Month-Year):	June-2022
Last Reviewed on (Month-Year):	-
Next Review on (Month-Year):	June-2025